

HERMANN PAUWELLS RIVERA

Software Engineer | AI/ML · Full-Stack

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PROFILE

Software engineer focused on AI/ML: Python (TensorFlow, FastAPI) and full-stack TypeScript (React/Next.js, Node.js); final semester of a B.S. in Computer Science and Technology at Tecnológico de Monterrey, available full-time from January 2027. Founder of three early-stage startups built around LLMs: AI-agent coordination, nutrition-label analysis and automated video production. Neural-network models for business cases (Datathon 2025).

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Kotlin, C++
AI & ML: LLMs (OpenRouter), TensorFlow, faster-whisper, Q-learning, AgentPy
Backend & Data: FastAPI, Flask, Node.js, PostgreSQL, Supabase, SQLite
Frontend & Tools: React, Next.js, Vite, Git, GitHub Actions, FFmpeg, Unity

EXPERIENCE

Full-Stack Developer (freelance), Juguetería El Arbolito Jun 2026 to present

- Built and shipped to production the online store and admin panel for a single-branch toy store with Next.js, TypeScript and Supabase (catalog, cart, checkout, backend).
- Integrated Mercado Pago payments and an LLM chatbot (OpenRouter); prepared the load of the 2,395-product catalog exported from the point of sale (Elevanta) and the online/in-store inventory sync (in progress).

STARTUPS (FOUNDER)

HIVEMIND Co-founder · team chat for people and AI agents · private pilot Sep 2026 to present

- Android (Kotlin/Jetpack Compose) and Windows (Electron) apps over a self-hosted ntfy server; HMAC-SHA256 signed messages and beta invitations with single-use codes.
- Python dispatcher that assigns each message or task to an AI agent using an LLM via OpenRouter (under US\$0.0001 per message), with primary/standby failover; 104 automated tests in the Windows client and an in-house security audit.

recaps-studio Co-founder · AI narrated-video factory (novel → manhwa-style video) Sep 2026 to present

- Resumable Python pipeline with hash-based caching in SQLite: script, TTS with word-level timing (faster-whisper), shot planning, AI images with a vision judge, subtitles and FFmpeg/Remotion rendering.
- Run modes with human approval gates, a credit budget cap and automated QA; tested with pytest and mocked providers so tests spend no credits.

Chécalo Founder · nutrition app for Mexico · in development Sep 2026 to present

- Scans barcodes (Open Food Facts) and computes Mexico's NOM-051 warning labels with deterministic rules checked against the official gazette (DOF); JavaScript PWA and Android app (Kotlin, CameraX, ML Kit).
- An LLM only explains the verdict, behind a proxy with a per-product shared cache, so AI cost grows with the catalog, not with users. Business model and unit costs documented.

SELECTED PROJECTS

CrediFiel Challenge (Datathon 2025) Data / ML · team project 2025

- Neural-network models (regression + classification) to optimize Credifiel's direct-debit collections: minimize commission costs or maximize recoveries.
- Real financial dataset of direct-debit collections and efficiency metrics (KPIs); Flask API to serve inferences and a React + Vite frontend.

Multi-agent warehouse simulation (RL) Python · AgentPy · Unity · FastAPI 2024

- Multi-agent Q-learning with conflict resolution, a blackboard system and task auctions.
- Animated Unity visualization and a FastAPI server.

Other projects: Lumina Reservations (WorkHub MTY) · folk-park · HeatShield · MAWI · VisualControl (IoT) · Mildred Pierce · GateGenius (HackMTY 2025).

EDUCATION

B.S. in Computer Science and Technology (ITC), Tecnológico de Monterrey, Monterrey Campus 2021 to 2026
Expected graduation: December 2026.

AWARDS & ACTIVITIES

- Winner, Sustainability category, at Swift Challenge Fest 2025 (Tecnológico de Monterrey), with the HeatShield iOS app.
- SwiftTec Coding Community (2025): hackathons and innovation challenges.
- Music production and sound design (FL Studio).

Spanish (native) · English (C1)